Computational Implementation of the Surya Siddhanta Across Fourteen Chapters

MTech Project Report

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Abstract

This project converts the computable procedures of the Sūrya Siddhānta into a Python library and makes them available through calculation, reading and conversational interfaces. The completed calculation layer contains one module for each of the text's fourteen chapters. Its scope includes mean motions, the sine table, true planetary positions, place and time, lunar and solar eclipses, eclipse projection, conjunctions, fixed stars, heliacal phenomena, the lunar crescent, pātas, cosmography, astronomical instruments and the reckoning of time.

The implementation follows Burgess's English translation and uses worked examples from Mahavir Prasad Srivastava's Hindi *Vigyan Bhashya* as checks. Tests tied to verses and examples have exposed errors that ordinary execution did not, especially in the interpretation of inner-planet motion, the sign of the *manda* anomaly and the Ujjain time base. A separate DE440 comparison measures the longitude path against a modern reference under specified conventions. It is treated as an evaluation of this implementation, not a proof of historical practice. The software, evidence for its behavior and remaining limits are examined together. The completed React application now also includes learning paths for the Sun, Moon, five planets and Rahu, alongside a Research Explorer. The lessons connect a date to correction stages and source references. The explorer makes the saved eight-body DE440 comparison inspectable across time windows and time treatments, with downloadable evidence. These additions improve access to the implemented model; no measured improvement in student learning is claimed.

Contents and abbreviations

Contents: Abstract 2; Introduction 4; Project scope and completed work 5; Source texts 6; Time conventions 7; Architecture 8; Chapters 1–14 9–19; Public interfaces 20; Corpus and chatbot 21; Testing and case studies 22–23; DE440 evaluation 24; Mars Learning Lab 25; Learning paths 26; Lunar nodes 27; Research Explorer 28; Interface validation 29; Limitations 30; Conclusion and references 31. The abbreviation SS refers to the Sūrya Siddhānta. JD is Julian Day, a continuous numerical date representation. Ahargaṇa is the elapsed day count used by the text. Jyā is the tabulated sine. Manda and śīghra name the two principal corrections to mean planetary position. DE440 is the Jet Propulsion Laboratory's modern planetary and lunar ephemeris used as a reference in a separate study. MCP is the Model Context Protocol through which an external model client can call the project's tools.

Introduction

The *Sūrya Siddhānta* gives a connected set of rules for astronomical calculation. Its fourteen chapters begin with measures of time and mean planetary motions, then proceed through true positions, local geometry, eclipses, conjunctions, stars, visibility, the lunar crescent, instruments and calendar reckoning. A verbal account can leave open which quantity a verse denotes or which clock a calculation uses. A working program cannot. Every ambiguity has to become an explicit choice, and that choice appears in the output.

The objective of this project is to make a traceable implementation of those procedures and test it against both the text and numerical examples. The calculation library is the centre of the work. The browser applications and conversational assistant are access layers: they let a user ask for a position, inspect a verse or explore a diagram, while chapter functions perform the calculations. A source passage, intermediate value and automated check make the answer easier to question and correct.

The software is both an astronomical calculator and a record of interpretation. A function that returns a longitude has to identify the epoch and the meaning of longitude; a function that returns an eclipse must also identify the observer. The same principle applies to explanations generated from the corpus. They should point back to a chapter and verse so a reader can inspect the adopted rule. The path from source to function to test leads to the application that presents the result.

The work has two audiences. A programmer needs a stable function and a defined input convention. A reader of the Sanskrit astronomical tradition needs to know whether an output follows a verse, a translator's commentary or a software convention. The same result can serve both only if provenance travels with the calculation. That is why the project gives chapter modules descriptive names, records verse references in code and keeps the corpus browser close to the calculation routes. The interfaces follow those design decisions and remain connected to the text.

Project scope and completed work

The implementation, interfaces and evaluation described in this report were carried out within the current project batch. This project brings together the Sūrya Siddhānta calculation library, its applications and its numerical evaluation as one body of work. Its objectives are to implement the text's computable procedures in reusable software, make the adopted rules and intermediate values inspectable, and evaluate the resulting longitude calculations against a stated modern reference. The report presents the full scope and completed deliverables together.

The calculation library covers the computable core of all fourteen chapters. The repository includes a chapter module for each chapter, a per-planet interface, panchanga functions, REST routes, browser interfaces, a tool-using chatbot and an MCP server. Verse-anchored tests, worked-example checks and a DE440 longitude comparison address distinct questions: whether a rule was transcribed consistently, whether it reproduces a printed example and how the implementation compares with modern positions. These checks should not be collapsed into a single claim of accuracy.

The chapter structure organizes the completed computational work. Chapters 1–3 cover mean motions, true positions, direction, place and time. Chapters 4–6 cover eclipses and their projected geometry. Chapters 7–11 address conjunctions, stars, heliacal phenomena, the Moon and pātas. Chapters 12–14 cover cosmography, instruments and time reckoning. The applications let a user browse verses, run a calculation through a web route, or call a tool through the chatbot or MCP server. The adopted constants are explicit and are not silently combined with a modern ephemeris.

The completed deliverables also include eight guided learning paths, a Research Explorer and a bundled Burgess corpus containing all 500 numbered verses across fourteen chapters. The true-place module implements the manda/śīghra correction sequence, and the chatbot's tool registry spans the chapter modules. Tests and research manuscripts document the computational edition and its numerical evaluation. Remaining algorithmic gaps, OCR noise and questions requiring independent review are stated as limitations of the delivered work.

Source texts and editorial approach

The principal source of rules and constants is Ebenezer Burgess's 1860 English translation of the *Sūrya Siddhānta*. The project also uses worked examples in Mahavir Prasad Srivastava's Hindi *Sūrya Siddhānta: Vigyan Bhashya*. Those examples print intermediate values for specific dates and places, making them particularly useful for checking signs, units and order of operations. The Hindi material has been consulted through OCR, so a doubtful digit must be checked against the page image. OCR can confuse the period typeface's nine and six; a disagreement is not automatically a software defect.

The code identifies where a procedure follows the verse and where commentary supplies an otherwise missing step. This distinction matters in Chapter 13 instruments and other passages where a description is not a complete numerical algorithm. The project implements a defensible reading of the computable core, not a critical edition of the Sanskrit text and not every historical practitioner's method.

Burgess's translation is not treated as a ready-made computer program. A verse may prescribe a relation but leave the starting value, order of correction or instrument construction implicit. Burgess's notes sometimes supply a method; where the project follows that reconstruction, it records the source. The second text supplies worked arithmetic, not a reason to erase a difference between the verse and commentary. This separation helps explain a mismatch without changing a historically motivated function merely to agree with one printed table. It also gives later readers a clear place to revise an interpretation.

Source attribution is especially important where a printed translation, its notes and a later commentary offer different amounts of detail. The project's implementation manuscript classifies disagreements with worked examples as a software defect, an intentional deviation, a source problem or a rule not yet implemented. This scheme prevents a test suite from being made artificially green by changing the historical rule to fit every printed number. It also prevents the opposite mistake of dismissing every mismatch as a misprint. The category itself needs evidence: a page image for a suspected OCR error, a verse reference for a rule, and an intermediate trace for a calculation.

Time and coordinate conventions

Julian Day is the shared numerical date representation throughout the library. The historical day count is anchored to the Kali epoch at the meridian through Ujjain. The project distinguishes a date interpreted in Ujjain local mean time from one interpreted in Universal Time. Passing the former to a sidereal-time routine as if it were the latter rotates local sky geometry by about 76 degrees. Ascendants, parallax, rising and setting and the armillary sphere all depend on this convention. An internally consistent function can still return a wrong result when the caller supplies the wrong clock.

The text's longitudes are sidereal. The implementation provides its own *ayaṇāṁśa* to obtain a tropical value when a comparison calls for one. Location-dependent functions accept latitude and longitude, but the reference meridian in the historical model is not Greenwich. State the date, local clock and longitude convention when presenting any numerical example.

The time convention affects tests as well as user input. A modern interface may receive a civil clock time with a UTC offset, whereas the chapter functions reckon from the historical meridian. The chatbot tools convert at this boundary and label the resulting zone. This avoids making each chapter function guess whether a fractional day means local civil time, Ujjain mean time or Universal Time. The DE440 evaluation has a separate reference clock, Terrestrial Time, and therefore computes a conversion involving the Ujjain longitude and ΔT. Reporting these clocks explicitly prevents a time shift from being mistaken for a flaw in planetary theory.

The historical longitude is measured from a sidereal reference. A modern DE440 position begins as a vector in a different frame and becomes a longitude only after a stated transformation. The project's evaluation rotates geometric geocentric vectors to the mean ecliptic and equinox of date, then compares circular angles. The *ayaṇāṁśa* used on the historical side is itself a model with a back-and-forth trepidation. A residual consequently contains both planetary-model difference and the chosen precession model. Reporting the raw sidereal value and the converted tropical value separately makes this visible to users of the Python API and web tools.

Calculation architecture

The Python package has modules `ch01_mean_motions.py` through `ch14_reckoning_time.py`. A module corresponds to a chapter and exposes its quantities to other modules. Later computations call earlier positional and calendrical functions instead of copying formulas. This arrangement makes the dependency chain visible: a mean longitude from Chapter 1 enters a true longitude from Chapter 2, which can then enter an eclipse, conjunction, visibility calculation or panchanga element.

The `planets` package regroups chapter functions under Sun, Moon, Mars, Mercury, Jupiter, Venus, Saturn, Rahu and Ketu. It is a convenience interface, not a second planetary model. Ketu is placed opposite Rahu as an interface convention because the text does not compute it independently. The package metadata requires Python 3.9 or later and NumPy as the core runtime dependency. Plotting, web services and model access add separate dependencies at the application layer.

The FastAPI application includes calculation routers and book routers for individual chapters. This matches the package structure closely enough that a user-visible result can be traced to a chapter function. The chatbot tool registry also calls those functions and supplies schemas for the model. The MCP server generates its tool list from that same registry. This shared path limits duplicated astronomical logic, although interface formatting and input conversion still require their own tests. The project's architecture document and `api/main.py` show the current composition; older README diagrams omit some later modules and should not be used as the sole architecture source.

The chapter modules return structured results where a calculation has several intermediate parts. A planetary-position result, for example, can contain mean longitude, true longitude, *manda* and *śīghra* contributions, *ayaṇāṁśa*, sign and *nakṣatra*. The chatbot tool serializes selected fields to JSON, adding human-readable context while preserving the numeric values. The API schemas and React client then give a typed transport boundary. This design is helpful for verification: a failing total can be traced to an intermediate term rather than requiring a complete reimplementation of the chapter to find the difference.

Chapter 1 mean motions

Chapter 1 establishes measures of time, the elapsed day count (*ahargaṇa*) and mean revolutions over a *mahāyuga*. The implementation converts dates to Julian Day, computes the elapsed count and obtains a body's mean longitude from its revolution constant and elapsed fraction of the cycle. It also supplies calendar quantities reused by the panchanga and later chapters. These constants matter because a small rate difference accumulates across centuries.

The inner planets require a careful reading of the revolution table. Mercury's and Venus's listed revolutions describe their *śīghroccas*, while the mean planetary place is linked to the mean Sun. Treating those entries as ordinary mean planetary motion makes the later *śīghra* anomaly collapse. The code now represents the distinction explicitly, and a test based on the planets' limited elongation from the Sun guards it. This is a case where a domain property catches an error that syntax and type checks cannot.

Mean longitude is a cyclical quantity, so the elapsed count and revolution count have to be interpreted together and reduced to a circle. The code has separate constants for the seven classical bodies, the lunar node and apogee. The node moves retrograde and therefore needs a sign convention distinct from ordinary prograde motion. The project's evaluation paper tabulates the adopted revolution counts and their implied rates, which makes secular drift measurable. Those modern-rate comparisons are an evaluation of the chosen constants; they are not inputs to the historical calculation. Chapter 1 tests also cover calendar and unit quantities that later chapters reuse.

The epoch itself is a mathematical input, not merely a date printed in a preface. Chapter 1's elapsed-day count determines the phase of every mean motion. A one-day mistake in the count has a different angular effect for the Moon, Sun and slower planets. The code therefore includes date conversion functions and tests for epoch and calendar behavior. The project's later local-time correction showed that a correct day count can still be misused when a modern clock function interprets it on another meridian. Mean-motion and place-dependent tests have to meet at that boundary.

Chapter 2 true positions

Chapter 2 supplies the 24-entry *jyā* table at radius 3438 and the corrections that move a body from mean to true position. The implementation interpolates tabulated sines and applies the *manda* and *śīghra* corrections in the prescribed half and full sequence. The *manda* anomaly is measured with the sign convention of the text. Reversing it changes the equation of centre's sign and can make a computed solar position several degrees wrong while the calculation still runs normally.

Modern inverse trigonometric and square-root operations appear in the routines, so the project implements the text's model with explicit numerical conventions. It does not reproduce every rounding step of historical hand arithmetic. The longitude path is particularly important because the same positions feed later eclipse, conjunction and calendar calculations, and because it is the part evaluated against DE440.

For the five star-planets, the *śīghra* correction uses a second moving reference. The Sun supplies that reference for Mars, Jupiter and Saturn, while Mercury and Venus have their own *śīghrocca* motion. The project's `true_longitude_full` path applies the corrections in sequence instead of adding two independent fixed offsets. That sequence matters because the later correction uses a position already modified by an earlier step. The table of *jyā* values supplies the historical trigonometric base, but interpolation and inverse operations are numerical choices in the implementation. A comparison with a modern longitude therefore measures the complete adopted path.

The code also computes the Moon and node from their own mean motions. Rahu is an ascending lunar node and is represented by retrograde mean longitude; the text supplies no corresponding periodic equation. The API therefore should not describe Rahu's value as an observed point in the same sense as a planet's true place. Ketu is derived at the interface as the opposite point. These distinctions matter when evaluating the node against a modern osculating reference: a difference can reflect the historical mean-node model and the reference definition, as well as numerical implementation.

Chapter 3 direction place and time

Chapter 3 turns a celestial position into local quantities for an observer. The implemented functions cover direction, gnomon shadows, rising and setting and the *lagna*. These calculations combine position with latitude, longitude and the local clock. Their tests therefore must check more than a body's longitude. The same value of JD interpreted on different meridians changes sidereal time and hence the relation between the ecliptic and the horizon.

The Ujjain time-base correction was one of the project's most consequential fixes. Before it, code for later parallax and instrument calculations could appear self-consistent while using a sky rotated relative to the observer. A horizon, equator and ecliptic diagram clarifies why a planetary longitude alone cannot determine a local rise time or ascendant.

The gnomon calculation connects a measured shadow with solar altitude. The *lagna* locates the ecliptic point at the horizon, and rising or setting computations locate the horizon crossing of an object. These are related through the observer's local frame but are not interchangeable outputs. The tests for this module use known geometric relations and chapter examples. The current project guide contains a stale Chapter 3 sample calling functions that do not exist; `ch03_direction_place_time.py` is the source for the current public functions. The distinction between documentation and verified code matters in a computational edition.

The chapter's local calculations provide the bridge from an abstract celestial coordinate to an observable direction. A longitude identifies a point on the ecliptic, but a horizon crossing also depends on obliquity, local sidereal time and latitude. The project uses the text's obliquity convention in its historical computations. The implementation manuscript reports that mixing the two clocks changed local geometry so strongly that a solar-eclipse parallax example exposed the error. This makes a general lesson concrete: tests of astronomical software should include the reference frame and time base, not merely a date string and an expected angle.

Chapters 4 and 5 eclipses

Chapter 4 derives lunar eclipse circumstances from the Moon, Sun and shadow. Chapter 5 treats solar eclipses, which need observer-dependent corrections because the Moon's apparent position changes with location. The implementation includes solar parallax terms *lambana* and *nati*. It distinguishes a geocentric computation from one that uses the observer's place. A result should be reported with the chosen place and date convention, because an eclipse visible at one location need not have the same circumstances at another.

The implementation has acknowledged limits. Full three-pass *asakṛt* iteration from a worked commentary example is absent, and some disc-diameter helpers retain fixed mean values. These details can affect contact estimates. The chapter implementation is therefore a useful executable reading of the adopted procedures, while an event-timing accuracy claim would require a separate dedicated evaluation.

Eclipse calculations bring several scales together: the difference between solar and lunar longitude, the node's place, apparent diameters and, for a solar eclipse, local parallax. A small mistake in one quantity can shift a contact or change whether a local eclipse is predicted. Chapter 4 and Chapter 5 keep lunar and solar eclipse paths separate because Earth's shadow and the observer's topocentric Moon are different geometries. The worked-example suite is useful here because it compares intermediate quantities such as *lambana* and *nati* with printed values. The limitations named above bound the conclusions one may draw from agreement at one example date.

The lunar and solar cases also differ in how parallax enters. For a lunar eclipse, the Earth's shadow and Moon are central to the geometry. For a solar eclipse, the apparent Moon depends on the observer's position on Earth, so a local result needs latitude and longitude. The project's route and interface layers expose both kinds of result. Its own evaluation paper confines its DE440 statistics to planetary longitudes and explicitly excludes eclipse timing. That boundary should remain clear: a degree-scale longitude comparison over ten-day samples cannot validate minute-scale ingress and egress predictions.

Chapter 6 eclipse projection

Chapter 6 converts eclipse quantities into a projected geometry with contact points and diagram data. Separating the projection from the event calculation is useful for both testing and explanation. A numerical calculation can be checked before it is rendered, while a figure can make the geometry easier to understand. The project provides plotting support as an optional part of the software; users of the core numerical library do not need it for planetary longitudes.

An eclipse figure is interpretable when its lines and discs, date and observer are identified, together with whether it depicts a historical-model prediction or a modern reference. A diagram records the geometry the program drew and must be read with the adopted eclipse assumptions. The chapter is a distinct computational layer that turns earlier angular quantities into an observable configuration.

The projection module is particularly useful for explaining an eclipse result to a reader who does not know the traditional terms. Contact points, discs and relative paths provide a geometric check on signs and directions: a numerical magnitude alone may hide a reversed axis or misplaced shadow. The module can return diagram data separately from plotting, allowing the core calculation to be exercised in tests without a graphics backend. A projection figure can show the chapter result with a caption naming the calculation and its assumptions.

The projection output can support a useful validation step. The relative path of the eclipsing discs should agree in direction with the signs of the computed latitude and contact offsets. If a plotted path moves the wrong way, the defect may be in coordinate conversion or plotting rather than the event search. The code's separation of numerical geometry from optional Matplotlib drawing permits tests to focus on the underlying values. In an academic presentation, a projection figure can accompany a small table of the same contact quantities, letting the reader compare the diagram with numbers.

Chapters 7 and 8 conjunctions and stars

Chapter 7 searches for conjunctions between planets and between a planet and a junction star. Chapter 8 contains star and *nakṣatra* data, including functions that connect a longitude to an asterism and calculate star rise or set. A conjunction is an event defined through changing relative position, so the implementation must search over time rather than calculate only a single date's value. The catalogue layer provides named objects and their coordinates for these searches.

The coordinates of the junction stars are an interpretive issue. Chapter 8 uses an ecliptic reading of the catalogue, while Chapter 13 uses a polar-coordinate rule to place stellar hoops on the armillary sphere. The two readings give different declinations for some stars. The project preserves the distinction instead of forcing every use of the catalogue through one unexamined coordinate interpretation.

Chapter 8's catalogue also links computational values to textual names. The *nakṣatra* lookup divides the circle into asterisms and portions, while the fixed-star functions use catalogue coordinates. The computational-edition study derives the listed *dhruvaka* values from the shares given in the verses and records the small effect of catalogue rounding. In Chapter 13, a different construction is needed for stellar hoops. The fact that these readings can be run side by side is useful: they can be compared for a particular star and date without pretending that the historical coordinate question has already been settled.

The `nakshatra_catalog.json` data and star functions give the catalogue a defined software representation. Averse-based star entry can be inspected through the corpus browser while its coordinates enter a calculation. The computational-edition manuscript notes that the *dhruvaka* values for twenty-seven stars follow a rule based on each star's share and the accumulated minutes of preceding asterisms. An apparent small discrepancy for Revatī is explained by rounding in the stored catalogue. This is a useful example of how a tabulated historical value, a derivation and a machine representation can agree only to a stated precision.

Chapters 9 and 10 visibility and the Moon

Chapter 9 treats heliacal rising and setting: the return of a planet or star to visibility after it has been close to the Sun. This is not the same event as a geometric conjunction. The library therefore exposes dedicated functions for finding the next heliacal event. Chapter 10 addresses the Moon's elongation, illuminated fraction, phase name, crescent width, horn angle, moonrise and moonset. The calculations combine angular position with local geometry where necessary.

These chapters show how a shared positional core supports questions that require additional assumptions. A phase can be inferred from Sun–Moon elongation, while a rise or set time needs a place. A claim of naked-eye visibility is more demanding than a geometric calculation and should be phrased according to the particular rule implemented. The chapter modules make those boundaries visible to a caller.

The Moon functions illustrate several levels of output from the same Sun–Moon geometry. Elongation is an angle, illuminated fraction is a geometric consequence of phase, and a crescent's width and tilt depend on further relations. Moonrise and moonset add the observer. The project's guide gives an end-to-end example for these Chapter 10 calls, while tests exercise the implementation. For heliacal events, the code searches forward from a starting JD rather than requiring the user to guess the event date. This design is suitable for an exploratory interface, provided that the output is explained as the result of the adopted visibility criterion.

The applications make these outputs available in several forms. A positions page can show the Sun and Moon on a zodiac wheel; a Moon page can show phase and crescent geometry; a star page can use a sky map. These visual elements are explanatory views of the same chapter results. They also expose where an input is missing. A phase calculation can run for a date alone, while moonrise needs a site. The chatbot's context includes a user place and UTC offset when available, and its tools are instructed to state what location and clock they used.

Chapter 11 patas

Chapter 11 implements Vyatipāta and Vaidhṛta as *pāta* events associated with equal declinations. The calculation corrects lunar declination for latitude, searches for the event and estimates its beginning and end from the relevant motion. This chapter illustrates why names should not be used as the only definition of an algorithm. The *pāta* here differs from the daily panchanga *nitya-yoga* terms that share the names Vyatipāta and Vaidhṛta.

A verse-level check changed an earlier duration computation: the denominator is the difference of daily motions required by the passage, not simply the rate at which two already-computed declinations separate. This difference materially affects the duration. The current module, API routes and user-interface page make the event inspectable while tests preserve the adopted interpretation.

The Chapter 11 module is exposed through `find_patas` and `compute_pata_details` chatbot tools, calculation routes and a Patas page. That spread of interfaces is useful for checking that all outputs share the same meaning. The event has a true instant and an interval; the interval depends on the rate specified in the verse, so a mistaken denominator changes duration even if the central instant looks plausible. The chapter's separate name and tests also help prevent confusion with panchanga *yoga*, where a similarly named daily category is computed from the sum of solar and lunar longitudes.

The *pāta* calculation shows how a derived event can magnify a small definitional mistake. The centre of the event is found by a condition on declinations, but the start and end depend on how quickly the relevant quantities change. A rate based on the wrong motion makes an interval too short even if the central condition is still reached. The implementation changelog records the verse-based correction and the corresponding API and UI exposure. Its tests can separately check the declination relation at the true instant and the interval arithmetic. This separation makes the numerical result easier to audit.

Chapter 12 cosmography

Chapter 12 treats the Earth's globe, distances in the text's cosmic model and cycles of time. The software derives planetary orbit distances from the Moon's stated 324,000 yojanas and the assumption that each body covers the same linear distance in a yuga. It also implements the Earth's circumference, a latitude-adjusted circumference and a longitude difference correction needed elsewhere in the book. These are computations within the *Sūrya Siddhānta*'s model. They should not be described as modern measured distances.

The module also represents descriptive material through a verse-tagged catalogue. Keeping the numerical derivation apart from descriptive entries makes it possible for an interface to present the source without implying that every cosmographic statement is a contemporary astronomical measurement. The chapter is a useful example of how the project handles material that is partly procedural and partly explanatory.

The cosmography calculation reproduces stated orbit values from a common linear-distance assumption and the Chapter 1 revolution counts. The stated Moon orbit, shared rule and derived orbit values can be compared directly with units. The project changelog notes that the derived values reproduce the text's figures closely and connect the Sun orbit to a constant already used in the lunar-eclipse module. That connection is an internal consistency check on the edition. It does not turn the yojana values into empirical modern distances.

Chapter 12 also carries material that a web page can display without forcing it into a numeric calculation: creation order, cosmic time divisions and other descriptive entries. The data catalogue tags these entries by verse. The numerical functions cover Earth dimensions, orbit derivation and time regents. In the project changelog, the derived orbit values agree with the stated values to within about 1.3 yojanas, a precise internal check for the implemented rule. The figure is a result of arithmetic and rounding in the historical model; it should not be confused with uncertainty in modern physical distances.

Chapter 13 instruments

Chapter 13 describes instruments rather than only a sequence of planetary formulas. The implementation represents the *gola-yantra* as a parametric armillary sphere in an equatorial frame. Hoops for the equator, ecliptic, horizon, colures, solar paths and stars can be passed to a renderer. The React interface contains an interactive three-dimensional representation, while the underlying module stores the astronomical geometry independently of the graphics library.

The chapter also implements the clepsydra and time finding from the gnomon, wheel, arc and staff. Some of these procedures are supplied by Burgess's commentary rather than fully by the verses. The output therefore identifies its source category. This is an important historical safeguard: a user can distinguish an operation directly warranted by the text from a useful reconstruction that completes an underspecified description.

The armillary model stores a hoop by its plane and offset, so a renderer can construct a sphere without recomputing astronomy. It changes with place and moment: horizon orientation and the ecliptic point on it are local, while some reference circles remain fixed in the model frame. The interactive page makes this movement visible. The instruments section also demonstrates a methodological point. Where a verse names a device but does not give every step needed to calculate time from it, the software may use Burgess's commentary, but the result says so. This is a more faithful presentation of uncertainty than hiding the added rule.

The Chapter 13 model uses an equatorial coordinate frame whose north pole and equinox axes are fixed by convention. A hoop is a geometric object in that frame, and a page renderer can rotate the view without changing the underlying astronomical values. For Ujjain, the computational-edition manuscript reports 46 hoops in its model. It also states that the eastern-horizon ecliptic point is computed directly instead of interpolated between zodiac signs. This matters because interpolation could make the interactive sphere look smooth while giving a measurably wrong local ascendant. The implementation keeps geometry and display responsibilities separate.

Chapter 14 reckoning of time

Chapter 14 returns to calendrical questions. The implementation covers measures of time, solar ingress, months and seasons, and named Jovian years. These operations draw on earlier solar positions and the historical day count, so they inherit the date and meridian conventions already discussed. Their output often has a familiar calendar label, but the path to that label remains a calculation with a particular rule and epoch.

The chapter also includes alternative ways of naming a Jovian year. The verse rule and the mean-longitude method supplied in commentary remain distinct where both are exposed. That choice exemplifies the project's larger approach: preserve competing procedures with their provenance when reducing them to a single silent answer would conceal an interpretive decision.

The code supplies time-unit conversion through a catalogue of *mānas* and calculates solar ingress events for a Gregorian year. It can also return ayanas, equinox-related crossings and *ṣaḍaśītimukha* divisions. The Fathers' day calculation and Jupiter's named-year cycle have separate methods. The `barhaspatya_year` function lets a caller select a method; the method is therefore part of a meaningful result alongside the year name. Chapter 14 handles calendrical categories whose definitions depend on the textual source, rather than treating a familiar modern calendar label as self-explanatory.

The time-reckoning chapter's `solar_year` function is organized around the Sun's entry into Aries in a Gregorian year. Related functions return the ingress events and seasonal divisions, while `convert` handles numerical conversion between measures of time. The *pitrya_clock* function addresses the Fathers' day using the Sun–Moon relation. These are distinct routes through the chapter; they should not be summarized as a single generic "calendar converter." The named Jovian year further demonstrates the need to label method, because a mean-motion method in Burgess's discussion and a method tied to heliacal behavior need not return the same label at every date.

Public interfaces

The chapter modules and Python API allow direct use by a programmer. The Gradio explorer groups results into pages for positions, panchanga, eclipses, conjunctions, stars and later chapters. The separate daily panchanga application presents a narrower almanac view. Streamlit pages cover several calculation groups, and a React frontend calls FastAPI routes for both calculations and book browsing. The FastAPI entry point includes chapter-specific routers, corpus, glossary and chat routes.

All of these surfaces depend on the chapter layer. The architecture matters because independently reimplementing a formula for each interface would make disagreements difficult to diagnose. A screenshot or sample response establishes what a user can inspect when its inputs are identified. Correctness comes from tests and source comparison, rather than the appearance of the interface.

The React frontend includes dedicated pages for the later chapters, including Cosmography and Instruments, which are not mirrored in every older interface. Its verse-browser pages call book routes, while interactive SVG and three-dimensional components render astronomical geometry. The Streamlit and Gradio applications have their own page sets. This difference in presentation should be described accurately: the same core package underlies the applications, but not every surface exposes every capability. The REST layer gives a convenient way to inspect JSON outputs and keeps the user interface separate from chapter arithmetic. A simple architecture figure should place the chapter package at the bottom, API and tool layers above it, and applications at the top.

For a researcher, the applications serve different tasks. A book page lets one inspect a verse and nearby text. A calculation page gives a result for chosen inputs. A visual component can show a configuration that a table alone obscures. The API route can preserve the precise JSON for a test or another program. The project's frontend also includes glossary access, reducing the need to know each Sanskrit term before exploring a page. The interfaces therefore provide several ways into the same computation, but the source code and tests remain the authoritative record of what was calculated. Learning paths (/learn) reuse the Chapter 1 and 2 APIs; /research presents the saved evaluation data.

Corpus retrieval and chatbot

The chatbot can call calculation tools across all fourteen chapters and retrieval tools that return verse-indexed text. The agent prompt directs it to call a calculation tool for a number and to retrieve the relevant passage before quoting or paraphrasing the book. The retrieval layer returns identifiers and source metadata, making citations possible in a generated explanation. When the indexed SQLite corpus is unavailable or empty, the retrieval code can build a local FTS5 index from a JSON export.

The repository includes a bundled Burgess corpus containing all 500 numbered verses across fourteen chapters. Chat retrieval, corpus search and Verse Explorer use this export by default, while explicit JSON and database overrides remain supported. Repaired verse boundaries and source locators improve traceability, but untouched OCR can still contain transcription noise. Optional Vigyan Bhashya commentary remains a separate source. The hosted assistant requires credentials for its configured provider: OpenRouter, direct DeepSeek or a custom compatible endpoint. The MCP server exposes the same calculation and corpus tool registry over standard input and output, with the client providing its own model.

The retrieval functions include single-verse lookup, chapter ranges, search and chapter index. Search uses identifiers rather than returning anonymous text snippets, which gives the chatbot a stable way to cite a passage. The agent additionally distinguishes a Burgess verse from a result taken from Srivastava's commentary; it should not attribute a commentary explanation to the ancient text. The tool prompt instructs the model to take computed numbers from calculation tools, not from its memory or an OCR passage. This is an architectural safeguard, though not a guarantee of a correct final sentence. A complete demonstration records a question, the calculation tool called, the retrieved verse ID and the response.

The assistant prompt specifies a bounded tool loop and asks that numerical answers come from the registered calculation functions. It can retrieve Burgess verse records and, when available, Vigyan Bhashya commentary hits, labelling the latter separately. The retrieval layer uses an indexed SQLite database when one is supplied and populated; otherwise, it builds a local FTS5 index from the bundled JSON export. This allows the checked-in Burgess corpus to support citations without requiring the external OCR pipeline at runtime. Commentary availability and remaining OCR quality still bound the explanations the assistant can support.

Test strategy

Verification has three layers. Verse-anchored tests ask whether each function follows the adopted rule. Worked-example checks ask whether a chain of calculations reproduces an example at its printed precision. The DE440 comparison asks how far a particular longitude path departs from a modern reference under specified conventions. Each layer catches a different class of problem. A code test can pass even when the chosen reading of a verse is wrong; a modern comparison can be close by chance; a printed example may contain a typographical or OCR error.

The computational-edition manuscript reports 1,256 chapter and per-planet interface tests and 257 selected worked-example cases. It also states that the audit-case counts refer to an audit branch whose cases and exact revision have not yet been archived. These are reported project results, not a newly executed count from this checkout. A future fixed release can record the exact test command and revision alongside the audit cases.

The strongest tests combine numerical assertions with source-based properties. An elongation bound catches a semantically wrong inner-planet formula. A printed eclipse example tests intermediate parallax, which can catch a clock shift. Wrapper tests check that the per-planet entry points remain aligned with their chapter functions. The repository also includes API and chatbot tests, so the interface path can be checked independently of the chapter mathematics. A test count belongs to a particular date and revision because this repository is still being developed. The source-grounded description of test classes remains valid even when the count changes.

The test organization follows the chapter structure, with dedicated files for Chapters 1–14 and additional files for API, chatbot, corpus and per-planet interfaces. The computational-edition manuscript separates chapter tests from wrapper tests, and a recent DE440 revalidation ran a focused Chapter 1–2 suite of 263 tests. Those figures refer to particular checks and revisions, so they should not be added together as a new total. The important verification principle is that a unit test should encode a textual or geometric expectation, while a worked-example test should preserve the printed inputs and output precision.

Worked-example case studies

The inner-planet correction is a clear case of a program running with the wrong interpretation. Treating Mercury's and Venus's revolution counts as their own mean motion made the *śīghra* anomaly zero and placed Mercury 153 degrees from the Sun on one tested date. A bound on elongation exposed the mistake. Another check fixed the sign of the *manda* anomaly: reversing apsis minus planet changes the correction and can shift the Sun's computed place by several degrees.

A Kashi solar-eclipse example exposed the time-base error. With Ujjain local time and Universal Time mixed, first-pass longitude parallax came out at 3.83 *ghaṭikās*, while Srivastava printed 1.06. A consistent clock brought the computed value to 1.16. The remaining difference is informative because the commentary uses a more complete iteration and a different disc-size treatment. These cases justify keeping intermediate quantities and source rules visible, instead of judging the software by a final answer alone.

The case studies are valuable because each identifies a symptom and a cause. For Mercury and Venus, the symptom was an impossible elongation; the cause was confusing a *śīghrocca* revolution count with a mean planetary place. For *manda*, the symptom was a correction in the wrong direction; the cause was reversing the textual anomaly convention. For the Kashi eclipse, the symptom was a large parallax discrepancy; the cause was mixing clocks. In each case a more precise test was added after the reading changed. This makes the implementation a revisable account of the text, rather than a collection of isolated outputs.

Another case involves the lunar apparent diameter. An initially plausible scaling placed the largest disc at apogee; the verse-based calculation scales the mean diameter by the ratio of true to mean daily motion. The worked example's intermediate numbers exposed the reversal. Chapter 12 offers a different kind of check: orbit values derived from the shared-distance rule nearly reproduce the printed values. Chapter 14 tests can protect named seasons and years against off-by-one shifts. Together these cases show that verification is not one metric. It uses domain bounds, arithmetic examples and cross-chapter consistency where each is appropriate.

DE440 evaluation

A companion study compares Chapter 1–2 longitudes of the Sun, Moon, five visible planets and lunar node with JPL DE440. It samples 40,176 epochs at ten-day intervals from 1550 to 2649 CE and considers both a direct clock comparison and a Ujjain/ΔT-corrected variant. Independent revalidation reconstructed all 642,816 stored residual values against the local DE440 kernel to within about 0.00000055 degrees. That verifies the stored residual arithmetic under the stated frame and time conventions. It does not validate every textual choice, later-chapter event prediction or historical panchanga practice.

The comparison adds the project's ayanamsha to its sidereal longitude to define a tropical comparison. A scalar precession-corrected residual is a diagnostic on longitude, not a full three-dimensional fixed-frame rotation. The text's trepidation model reaches a maximum near 2299 CE and then reverses, affecting future extrapolation. The repository's revalidation note records concerns from an earlier manuscript snapshot; it must be read with its October 2026 review date and revision. The current repository also supplies regenerate_paper.py and REPRODUCE.md. The new explorer does not rerun that pipeline or independently reassess its fitted coefficients.

The sampled DE440 interval lies inside the local kernel's coverage. The study uses geometric geocentric vectors, converts them to a mean ecliptic and equinox of date, and compares wrapped longitude differences. The Moon's node uses a separate reference derived from its orbital angular momentum, so its residual differs physically from a planet's. Future epochs test extrapolation of the adopted constants and trepidation model; they are not observations already made. The current explorer groups the saved decimal-year samples into the same windows as the saved tables, with every exported group checked against those tables.

The earlier revalidation identified twelve epochs whose former decimal-year labels assigned them to the preceding century. That historical finding illustrates why residual arithmetic and later grouping need separate checks. The current explorer tests verify all 176 body-window-treatment groups against the current saved tables, along with source hashes, sample coverage and retained extrema. This establishes faithful presentation of the supplied evidence, not an independent rerun of the kernel or a validation of event timing. Angular elongation errors should not be converted directly into claims about eclipse or tithi timing without checking actual event crossings.

Mars Learning Lab

The /learn/mars route turns the Chapter 2 true-place calculation into a guided experiment for Mars. A whole-day input and a slider spanning 90 days on either side of the chosen date control the existing book API. Results are evaluated at midnight in Ujjain mean time and displayed as sidereal angular directions from Earth. The drawing shows angles, not physical orbital distances.

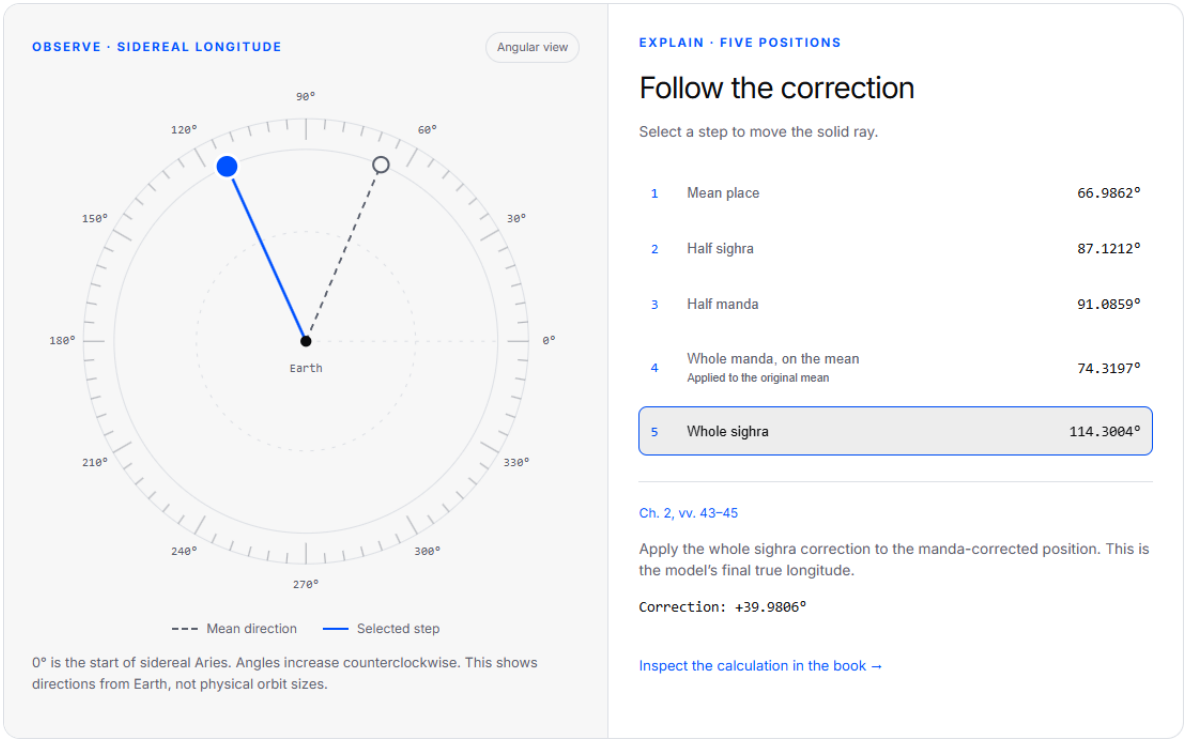


Figure 1 Mars Learning Lab showing the final correction stage for 4 November 2026

The five selectable positions are the mean place, half sighra, half manda, whole manda on the original mean, and whole sighra. The fourth step explicitly returns to the original mean, avoiding a misleading cumulative interpretation. At the demonstration date the final longitude is 114.3004 degrees. A link opens the source calculation with the same date; optional corpus text is distinguished from calculation availability.

An understanding check asks whether the final position is ahead of or behind the mean along the shortest angular separation. JSON export preserves the inputs, time basis, frame and intermediate results. This is implemented teaching functionality; its effect on student learning remains unmeasured.

Learning paths

The /learn directory now offers eight working lessons: Sun, Moon, Mars, Mercury, Venus, Jupiter, Saturn and Rahu. Learners can start with the Sun, then study the Moon before tracing the planetary corrections. The directory supports direct links to any lesson; it does not require a prescribed order or claim measured mastery.

Learning path	Implemented experiment
Sun	Mean and manda-corrected longitude; tropical Sun, declination and day/night length at Ujjain.
Moon	Two correction positions; Sun–Moon elongation, 12-degree tithi intervals and waxing/waning paksha.
Mars	The original five-position lesson and its saved research panel.
Mercury and Venus	Individual lessons and a same-date comparison of every correction stage and Sun separation.
Jupiter and Saturn	Individual lessons and a same-date comparison of the five-position sequence.
Rahu	Ascending and descending lunar nodes, regression, longitude geometry and an eclipse concept quiz.

All seven body lessons share date validation, a direction diagram, selectable API-derived stages, chapter references, a prediction check and JSON export. Dedicated panels connect the calculation to each topic. Sun and Moon have two positions rather than the five used for the star-planets. Planet comparisons use one API response so both bodies share the input date.

The seasonal panel explicitly changes from sidereal to tropical longitude and discloses the library's hour-angle day-length method. The Moon panel displays elongation and tithi without claiming to render the illuminated disc. Each tithi spans 12 degrees of longitude separation and need not last one civil day. Paired-planet Sun separation is a longitude difference, not an observational visibility estimate.

Downloads include the lesson's additional context: day-length values for the Sun, tithi results for the Moon, or comparison rows for paired planets. Late responses cannot overwrite a newer date. Invalid or failed requests keep exports disabled and label any previous result. Optional corpus failures leave the numerical lessons available.

Lunar nodes and eclipse geometry

The Rahu lesson at /learn/rahu draws the ascending lunar node returned by Chapter 1 and derives the opposite descending direction by adding 180 degrees modulo 360. The Sun and Moon positions come from Chapter 2 for the same input date. Learners can select either node or the Moon direction and observe the node's negative mean motion as regression.

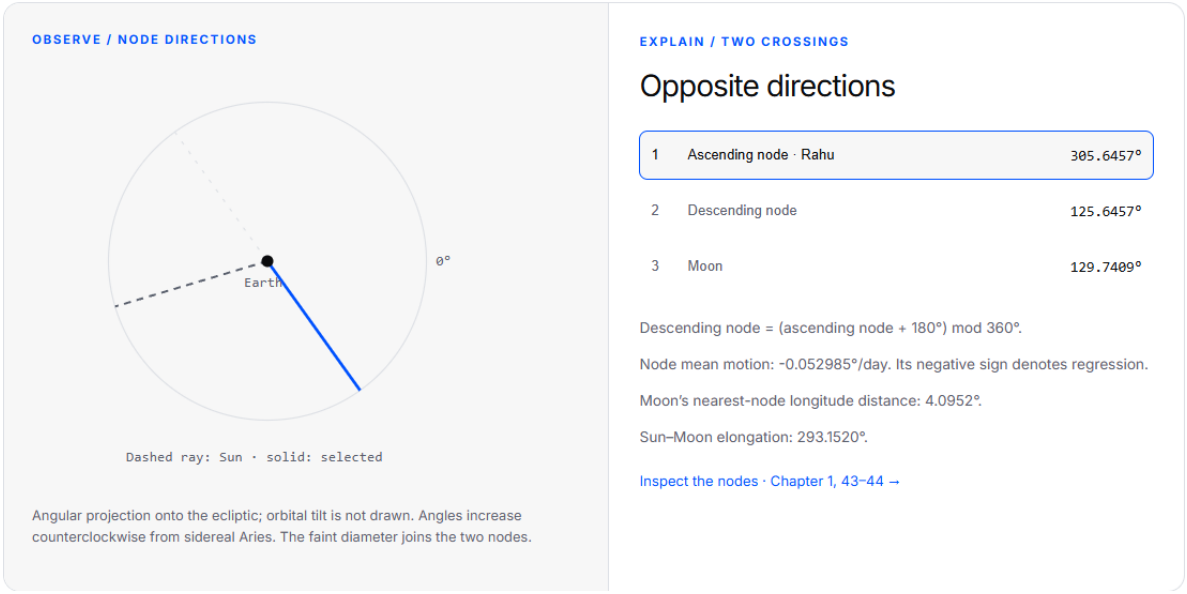


Figure 2 Rahu lesson showing the two lunar nodes and same-date Sun and Moon directions

The projection shows ecliptic longitude only, with orbital tilt described in text. The lesson explains that solar eclipses require new Moon near a node and lunar eclipses require full Moon near a node. Its concept check asks why phase alone is insufficient. The export records event_prediction as false. Event occurrence, contacts and local visibility require further geometry; links lead to the existing eclipse calculators without treating this diagram as an event prediction.

The conceptual background is supported by NASA Science, Eclipses and the Moon (https://science.nasa.gov/moon/eclipses/) and Moon Phases (https://science.nasa.gov/moon/moon-phases/), consulted 4 November 2026. Historical node and tithi references remain the project's Chapter 1 verses 43–44 and Chapter 2 verse 66. Modern explanatory references do not replace the adopted historical model.

Research Explorer

The /research route exposes the saved longitude study for the Sun, Moon, Mercury, Venus, Mars, Jupiter, Saturn and Rahu across eleven 100-year windows from 1550 to 2649 CE. An 8 by 11 RMS matrix supports direct selection, with native body and window selectors as an equivalent. Rahu is identified as the ascending lunar node. The two time treatments share a vertical scale within each selected comparison.



Figure 3 Time-aligned RMS longitude errors from the saved DE440 comparison

The naive treatment reads library JD directly as reference TT. The time-aligned treatment subtracts 75.7873/360 day for the meridian and adds the paper's approximate Delta-T in days. It changes the reference epoch, not the historical formulas. For the Moon in 1550 to 1649, the saved RMS changes from 3.990 to 1.686 degrees. A lower RMS in one window is a diagnostic result, not calibration of the SS model.

The chart preserves the minimum and maximum sampled residual in each five-year group and connects the group means. Full-window statistics use every original ten-day sample. JSON and CSV downloads include source paths and SHA-256 hashes, frame and time conventions, and sample counts. The explorer needs no live JPL request.

Interface validation and reproducibility

The repository verification record dated 8 November 2026 reports that PYTHONPATH=src python -m pytest -q passed 1,677 tests after the corpus repair. The learning-path validation record dated 4 November reports 1,653 passing tests, an earlier focused run of 49 tests across test_learning_lab_data.py, test_research_explorer_data.py and test_api_book_ch02.py, and a successful production frontend build. These are dated repository results rather than fresh test runs performed while editing this report.

Check	Evidence
Source-data integrity	All 176 body-window-treatment groups match the saved tables; each body has 40,176 samples per treatment.
Aggregation and provenance	Source hashes, complete sample counts, RMS, signed means and sampled extrema are checked; generated files must match fresh exports.
Browser workflows	Real API values, correction stages, whole-day stepping, date-preserving links, rapid responses, failure/retry, optional corpus, understanding feedback and exports.
Research interaction	88 matrix cells; all 16 body/treatment combinations; JSON and CSV metadata; keyboard selection and mobile overflow checks.
Visual review	Light and dark desktop layouts and a 390-pixel mobile layout; screenshot evidence retained with the project.

The reproducible frontend data exporters are docs/evaluation/export_learning_lab.py and export_research_explorer.py. They summarize saved CSVs; they do not compute fresh DE440 reference positions. The source-study pipeline remains separately documented in docs/evaluation/REPRODUCE.md.

Browser checks in frontend cover all eight lessons and the research explorer. They verify API values, lesson evidence exports, node geometry, quizzes, date-preserving source links, stale responses, retry and mobile layout. Missing corpus text leaves the calculation usable. These checks establish the tested software behavior; historical authenticity, eclipse timing accuracy and educational effectiveness require separate evidence.

Limitations and next work

The library covers the computable core of all fourteen chapters, not every verse. Remaining gaps listed in the computational-edition manuscript include full eclipse iteration, some disc-diameter and visibility rules, parts of the equation of time and descriptive passages that are not algorithms. The bundled Burgess corpus has complete numbered-verse coverage, while remaining OCR noise and the optional external commentary still require review. Star-coordinate interpretation and procedures completed from commentary are documented choices rather than uniquely settled historical facts. Modern trigonometric operations also mean the software does not simulate every rounding step of historical hand calculation.

The next work should archive the worked-example audit at a fixed code revision, independently rerun and review the current DE440 regeneration and fitting pipeline, and extend remaining procedures where a defensible rule can be specified. The new teaching and research interfaces make these questions visible, but do not resolve them. A learner study remains future work. The eight completed learning paths cover the Sun, Moon, five planets and Rahu, but the understanding checks alone do not establish educational effectiveness.

The text is partly computational, partly descriptive. Completing every verse as a function would require inventing a numerical rule for passages that do not provide one. The current project's claim is therefore narrower and more testable: it covers the computable core across all chapters, with listed exceptions. Improvements should proceed through explicit source decisions. For example, a full eclipse iteration can be added with tests against the commentary's intermediate passes; a revised star-coordinate reading can be compared against both current methods; and the available DE440 pipeline can be independently rerun to regenerate each table from a fixed kernel and code revision. The modular structure makes such changes reviewable.

The open issues are tractable because they have clear locations. The eclipse iteration belongs in the chapter-specific event functions and can be checked against Srivastava's intermediate passes. The bundled corpus needs continued OCR review and fixed-version provenance for future revisions. The DE440 study needs independent rerunning and review of the available regeneration and fitting pipeline, with a fixed code and kernel revision, so another researcher can assess its figures and fitted results. The star-coordinate alternatives need documented comparisons before either is declared definitive. Addressing these items would strengthen the project's scholarly use, because a user could then trace more conclusions from a stable source through executable code to a repeatable result.

Conclusion and references

This project delivers a chapter-wise calculation library covering the computable core of all fourteen chapters of the Sūrya Siddhānta, together with its interfaces, source retrieval and evaluation. Its strongest result is the record of how a textual rule becomes a calculation: a passage fixes a rule, a function executes it, and a test or worked example checks the consequence. This chain has corrected plausible but wrong numbers. The interfaces make the work accessible, while the DE440 comparison separately measures the adopted longitude path against a modern reference. Known limits define what the software currently supports. The completed learning paths and Research Explorer provide a demonstrable workflow: choose an input, inspect correction stages, follow the source reference and examine the comparison evidence. They use the shared calculation library and saved study, with software checks and explicit methodological limits.

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